

Background

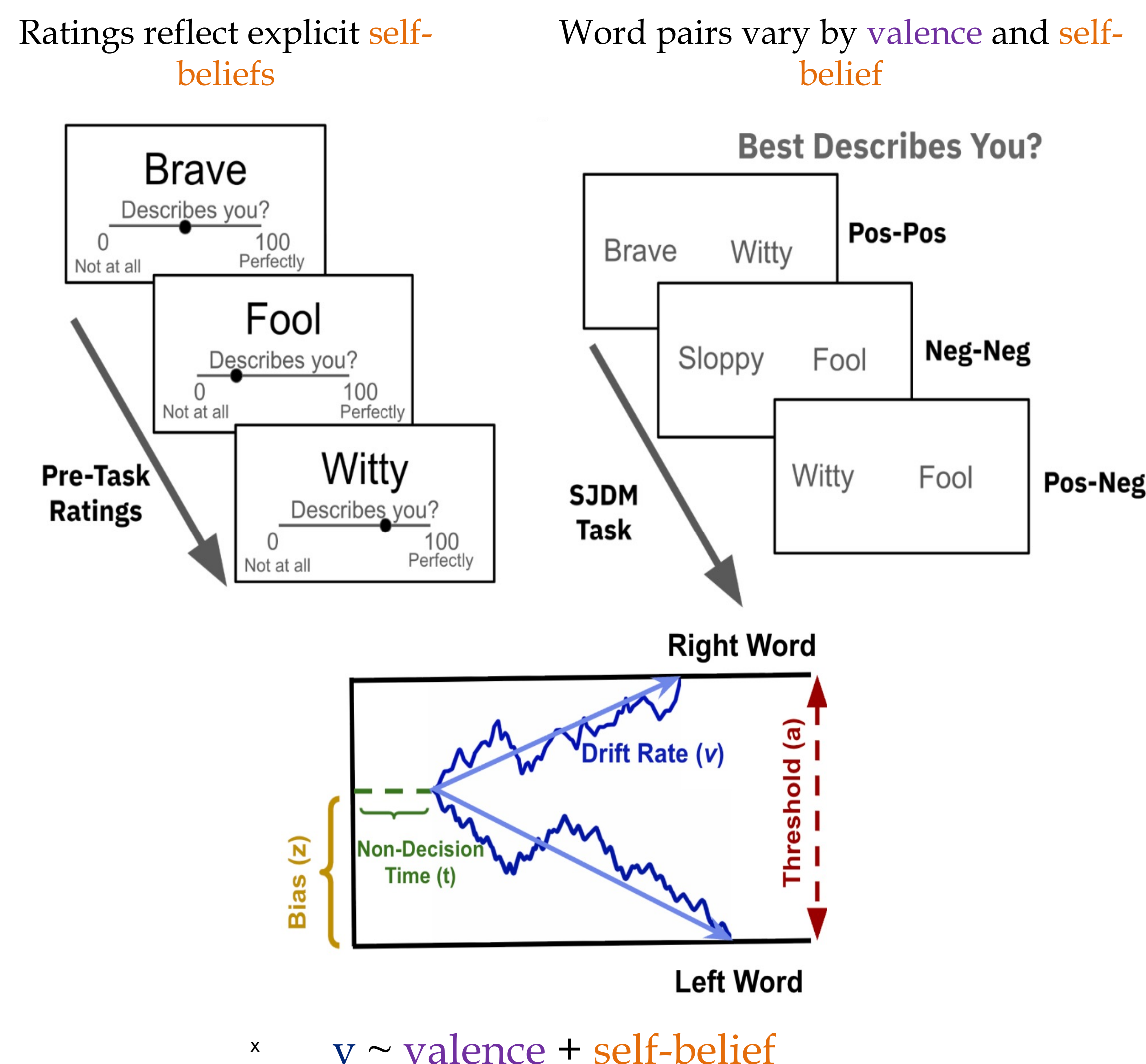
Motivation

- Enriched negative (vs. positive) self-schemas are a risk factor for depression.¹
- Decision-making models quantify this risk factor as a valence-dependent drift-rate ($v \sim \text{valence}$): the rate in which one accumulates (i.e., sequentially samples) negative relative to positive information about oneself.²
- Recent evidence suggests that when controlling for explicit self-beliefs, $v \sim \text{valence}$ varies across individuals as a function of depression.³
- The seminal mood congruency theory of depression states that low mood facilitates processing of negatively valenced self-referential information. However, this has not been formally tested.⁴

Task & Modeling Paradigm

Self-Judgment and Decision-Making Task (SJDM)

(Pandey et al., 2026)



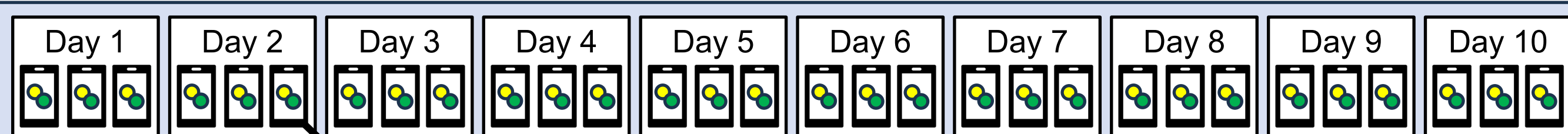
The present study seeks to mechanistically test the theory of mood congruency by:

- Validating a novel task and modeling paradigm for intensive longitudinal assessment.
- Assessing whether valenced self-referential processing ($v \sim \text{valence}$) varies within-person as a function of momentary affect.

Study Design

Baseline Visit
~1hr zoom visit

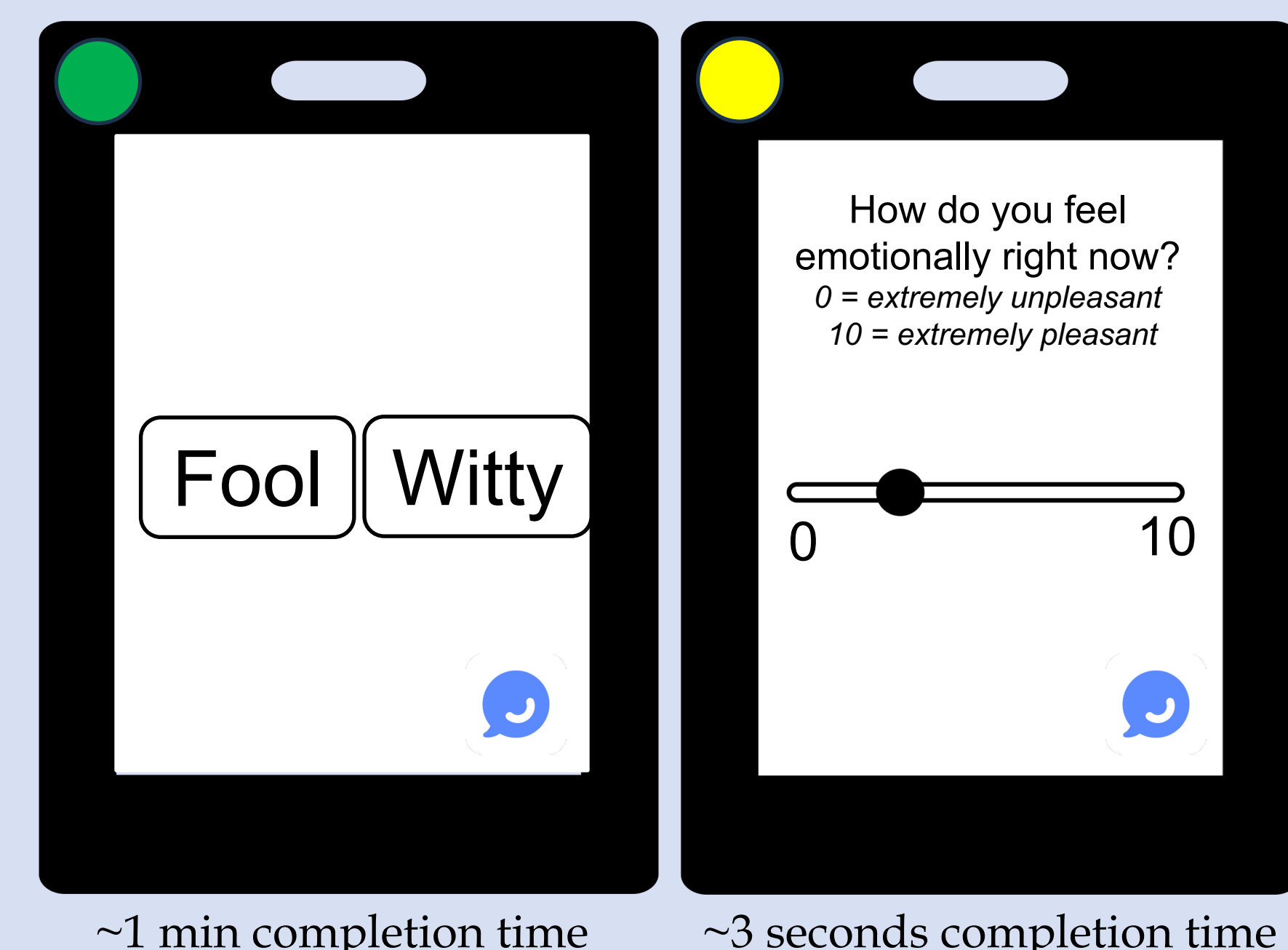
Ecological Momentary Assessment: Fabla Smartphone App
30 EMAs Total = 3x Daily for 10 days



Example EMA

- Self-report affect
- SJDM-EMA Task

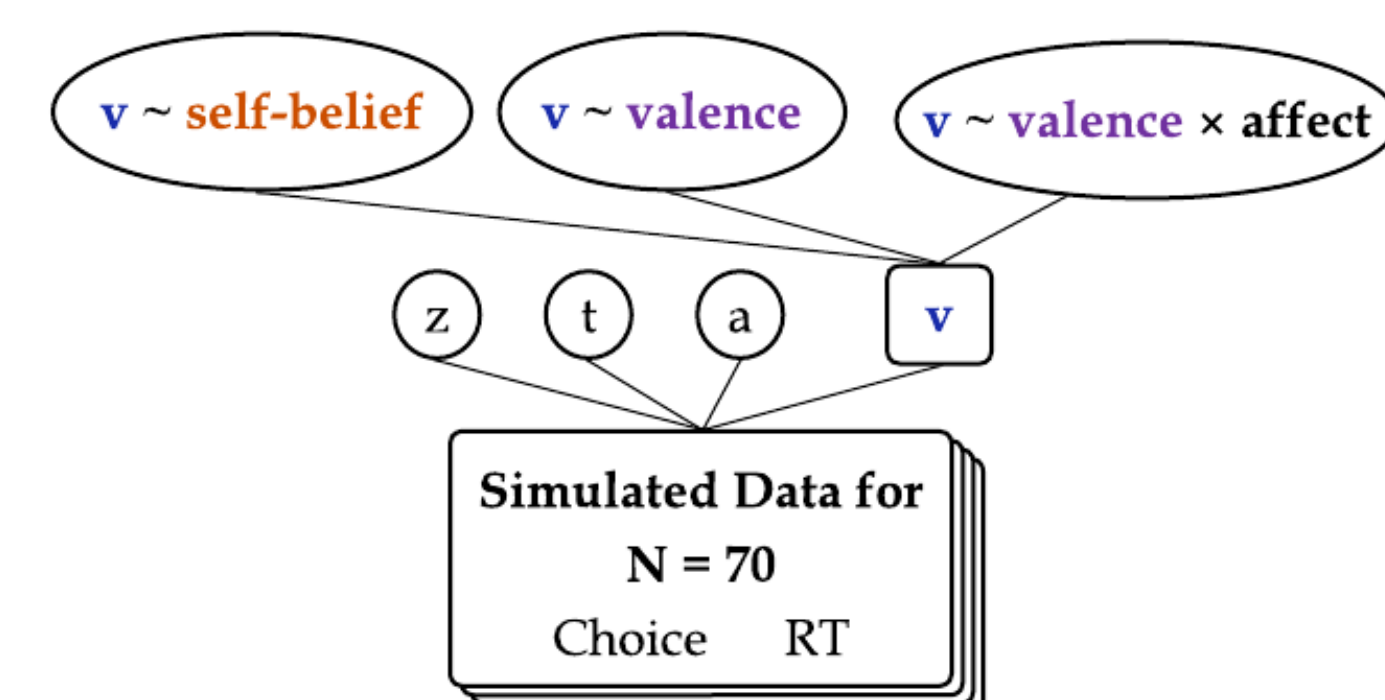
x 24 word-pairing trials in each EMA



- Rate 160 words
- Psychometric Questionnaires

Power Analysis

Data Simulation from Hierarchical Drift Diffusion Model



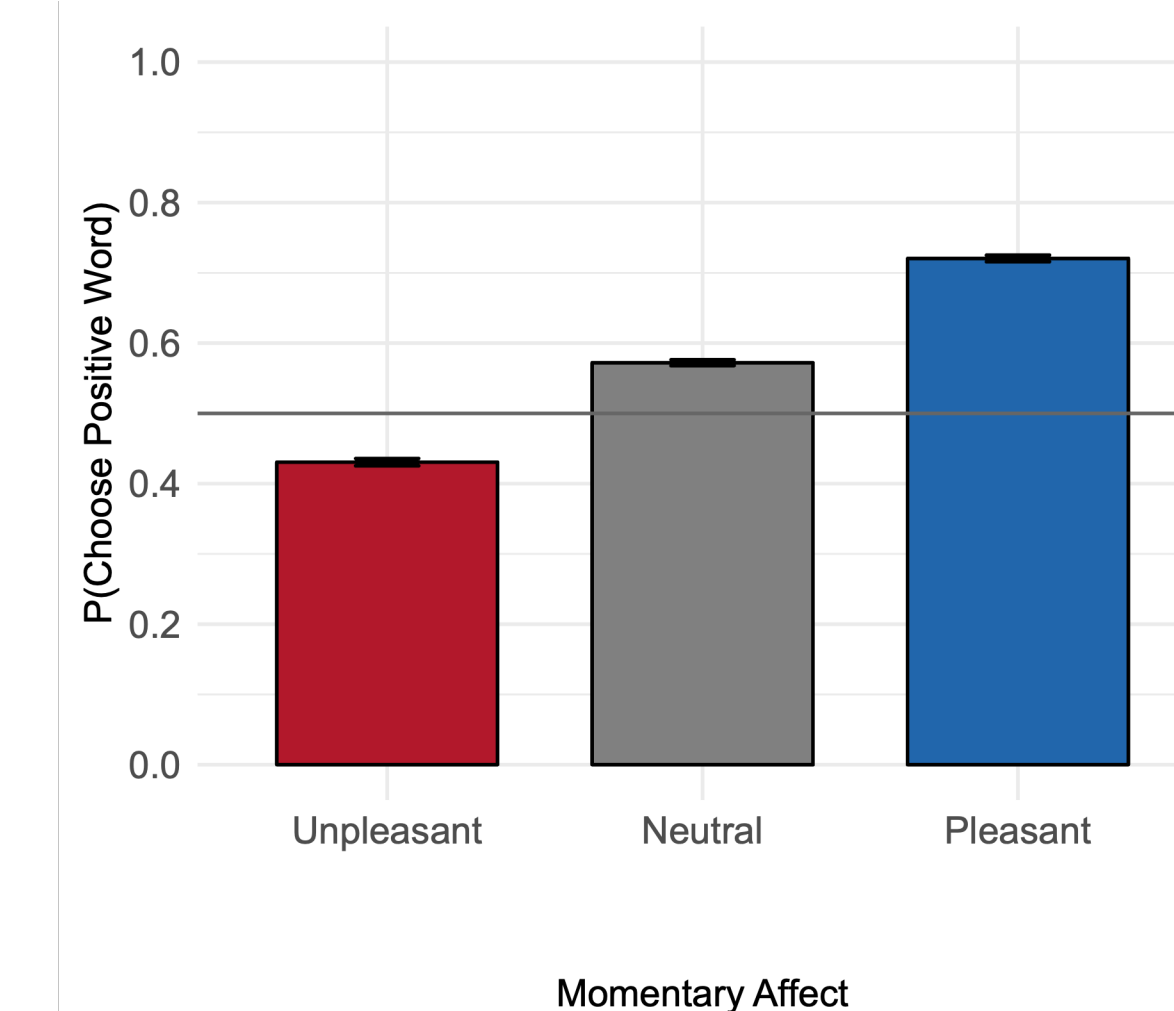
Mixed effects models detected predicted Choice and RT effects in all simulated experiments.

Study design and $N = 70$ target sample size provides adequate power to detect predicted effects.

Behavioral hypothesis 1.

Main effect of affect on Choice

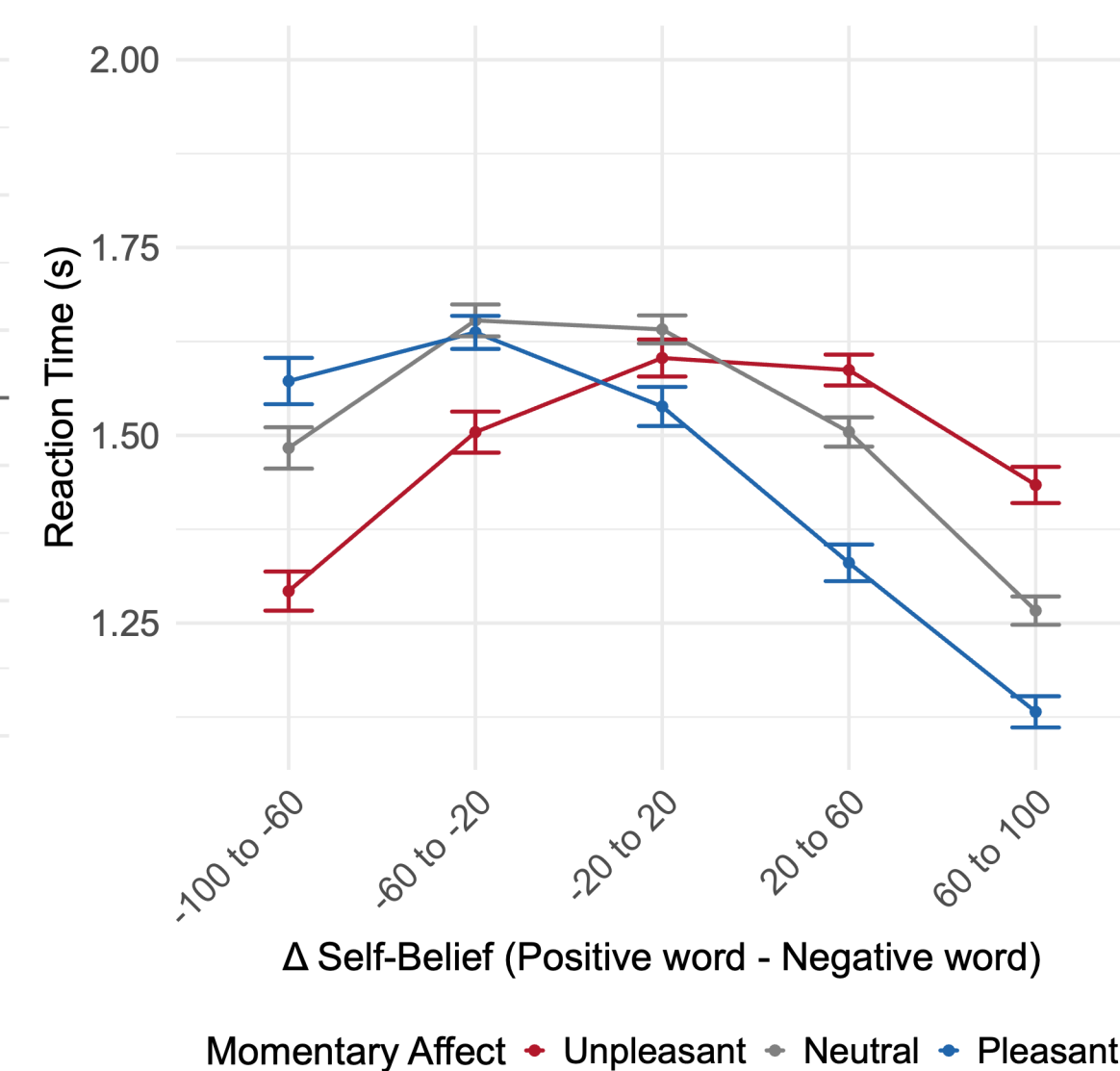
- When feeling more pleasant (unpleasant) participants will choose a higher (lower) proportion of positive words as self-descriptive than when affect is at its average value.



Behavioral hypothesis 2.

Interaction of affect and self-belief on RT

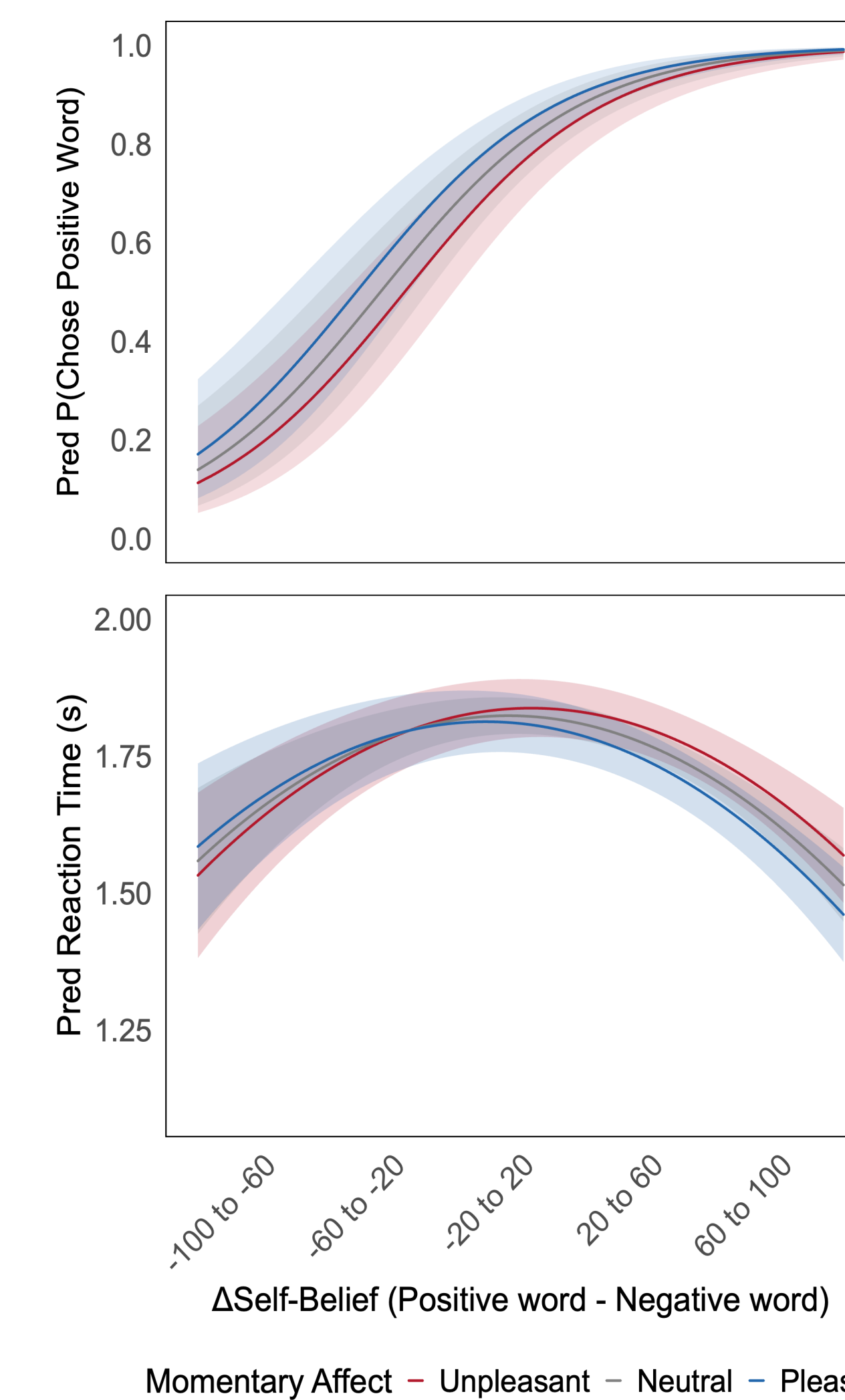
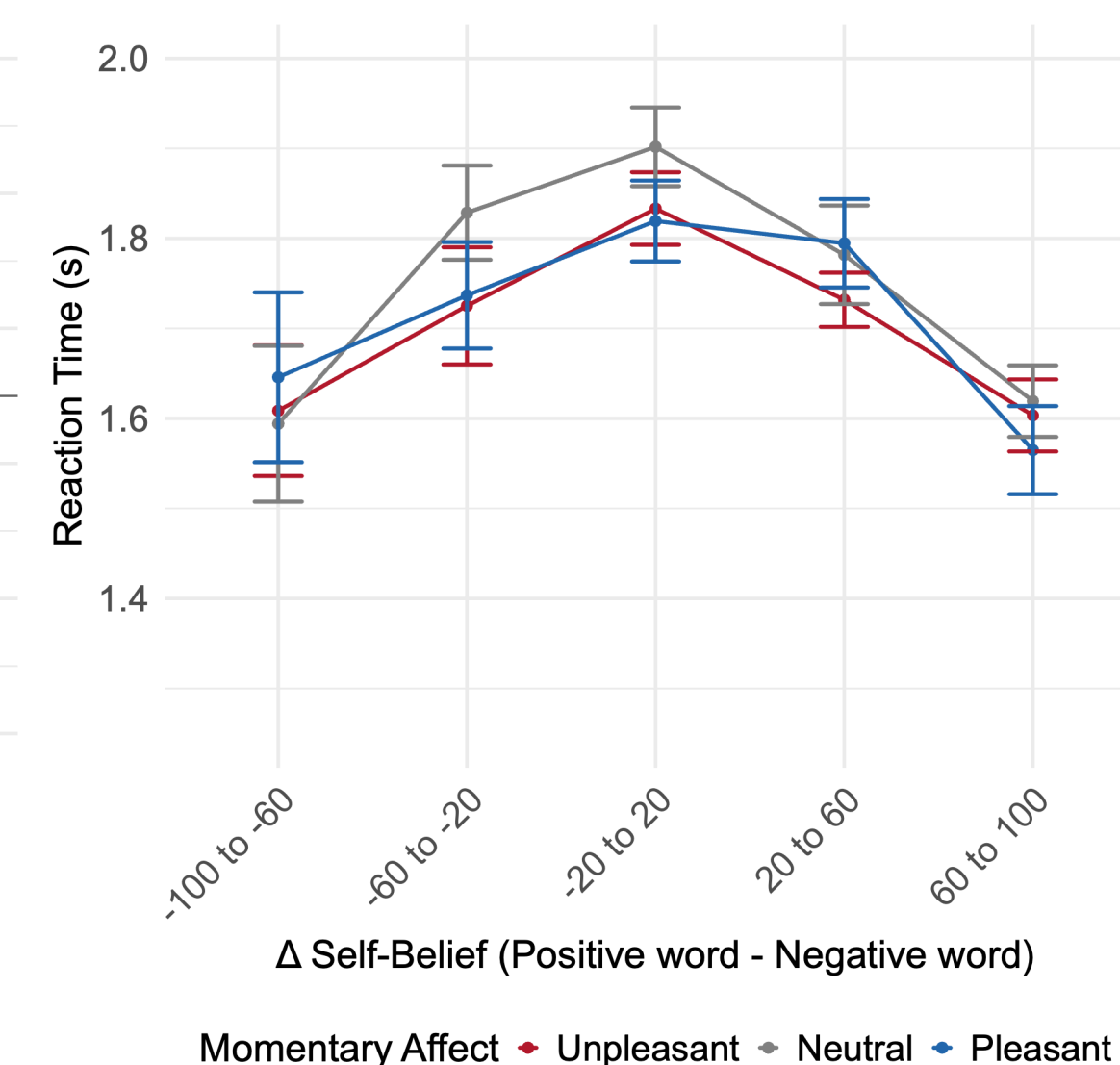
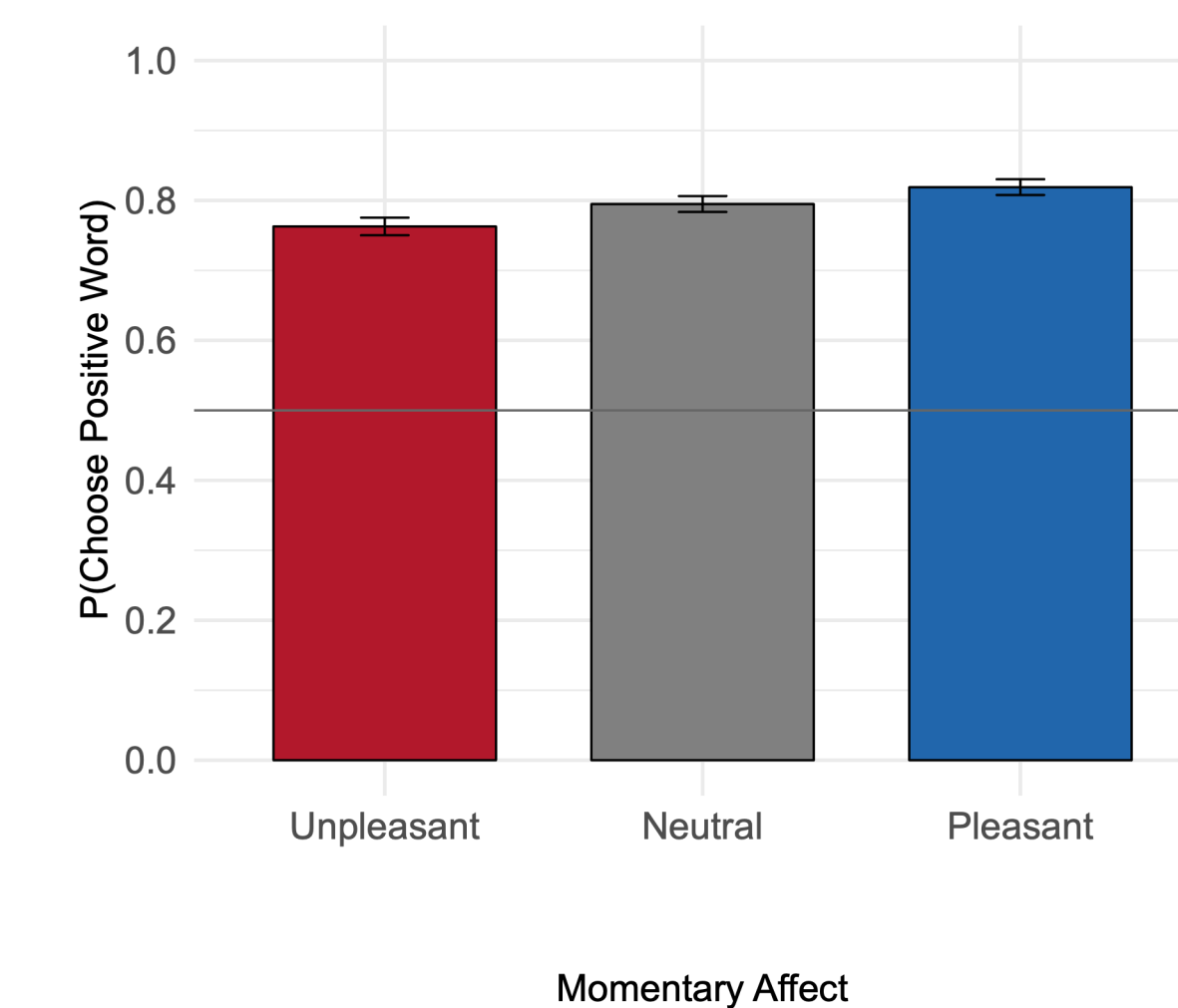
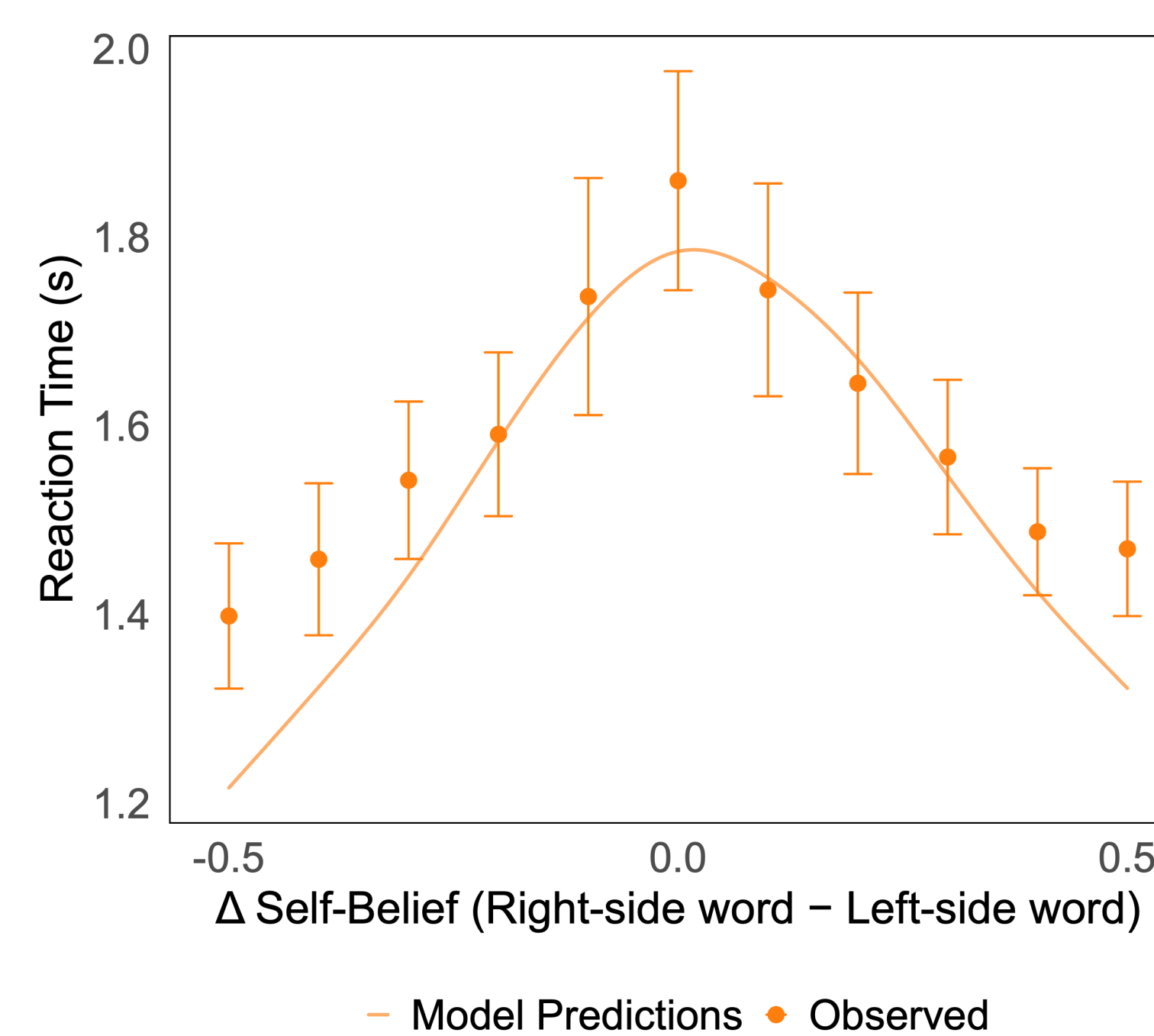
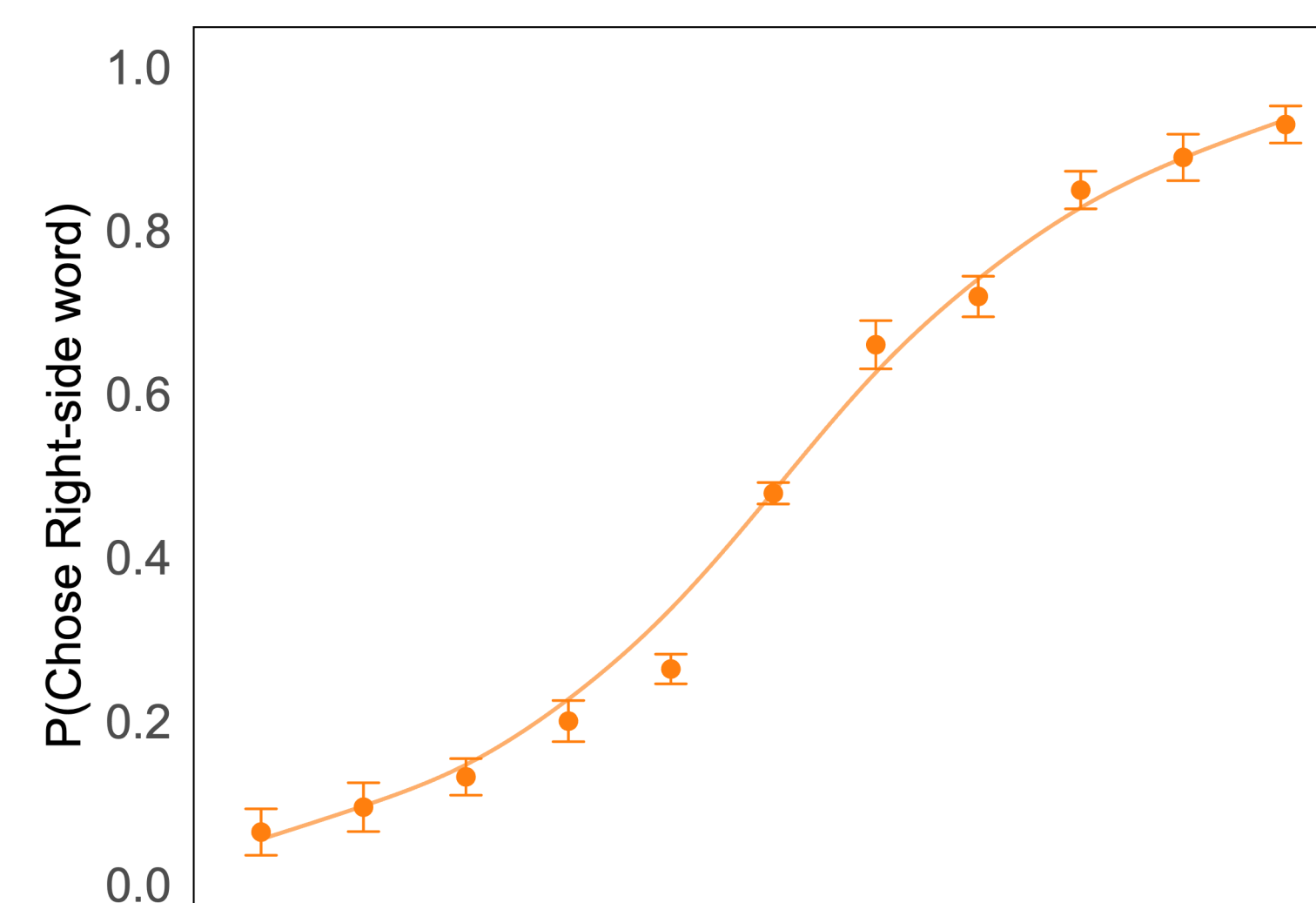
- When feeling more pleasant (unpleasant) participants will choose a higher (lower) proportion of positive words as self-descriptive than when affect is at its average value.



Preliminary Findings

$N = 26$ (out of $N = 70$ target sample) size have completed protocol.

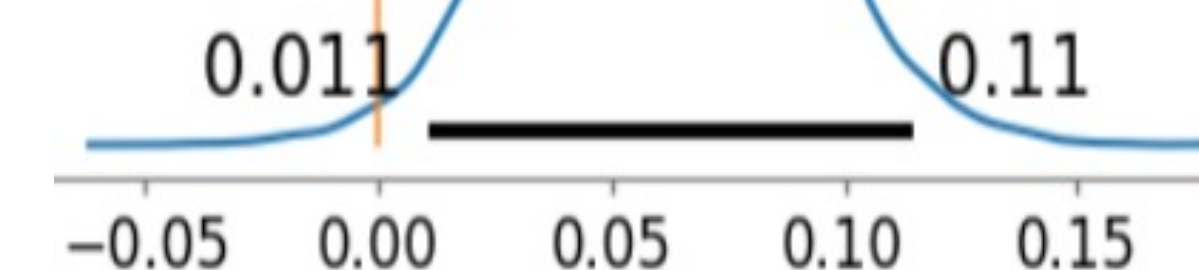
Preliminary support for mood-congruent shift in evidence accumulation for valence



$v \sim \text{valence} \times \text{affect}$

mean=0.061

95% HDI



Replicated canonical effects of evidence accumulation for self-beliefs in smartphone data

Scan for References + Supplementary Material + Analysis Code

